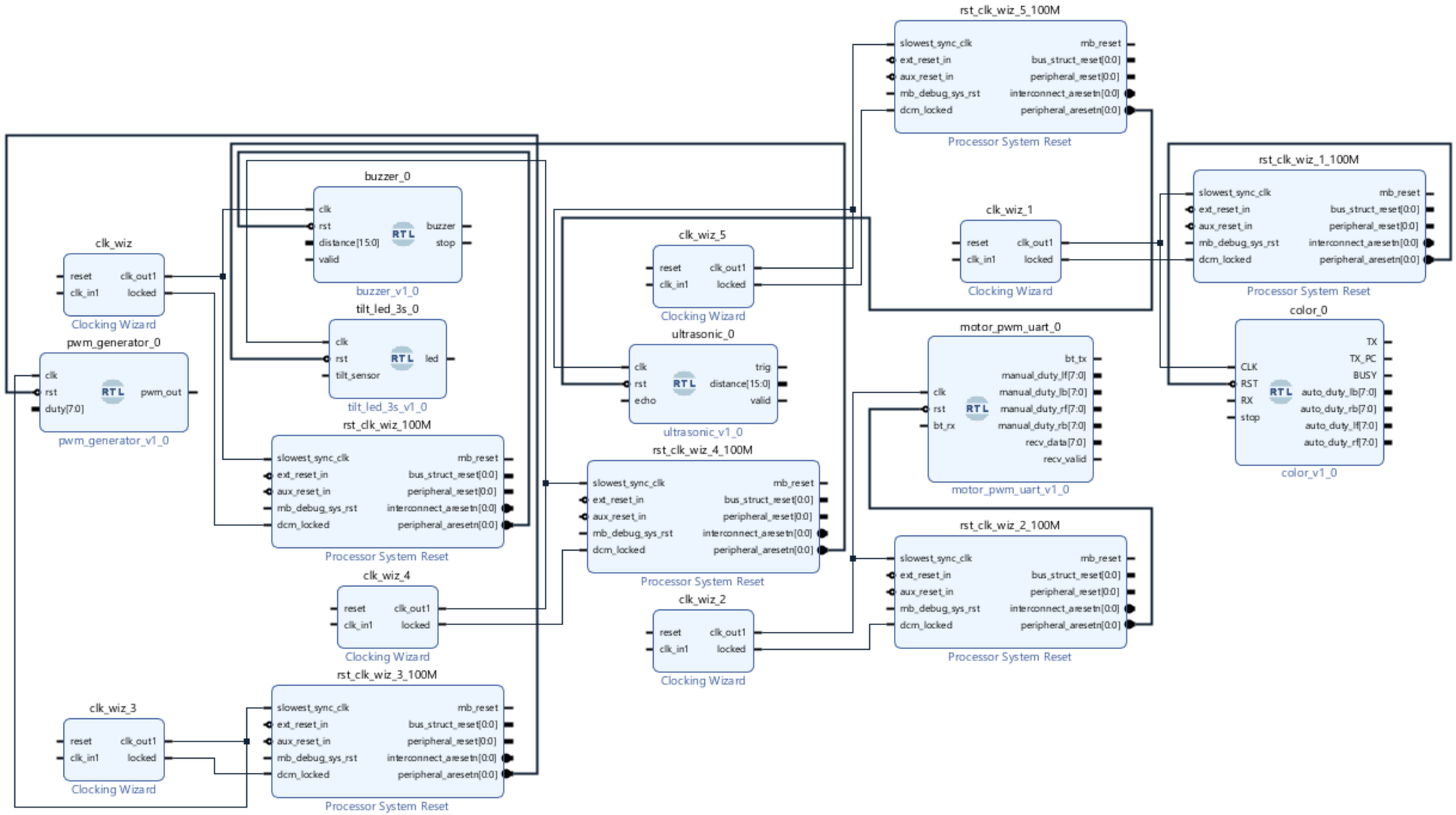
자율주행
RC카 제작

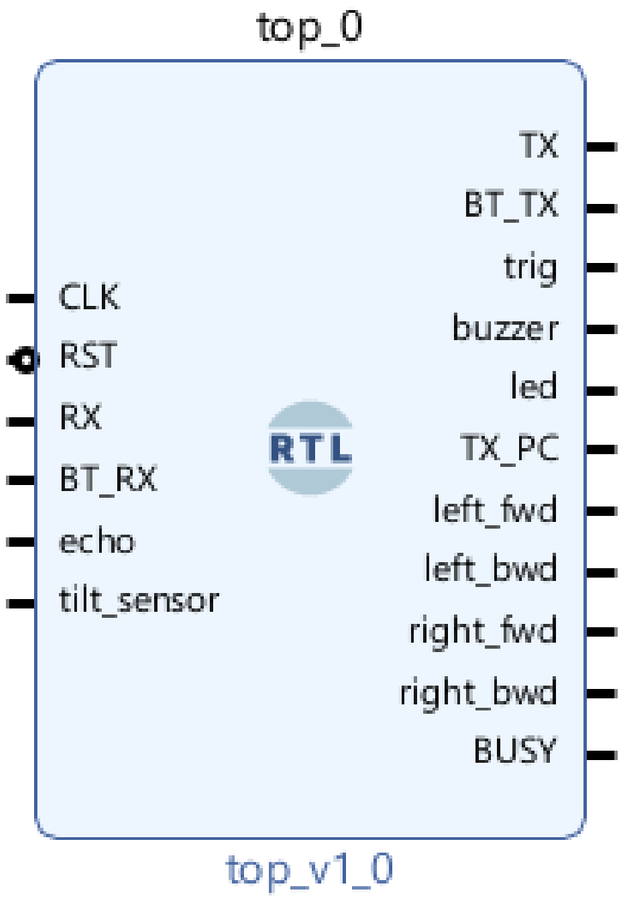
2025.5.22

6조 이지원, 정우석

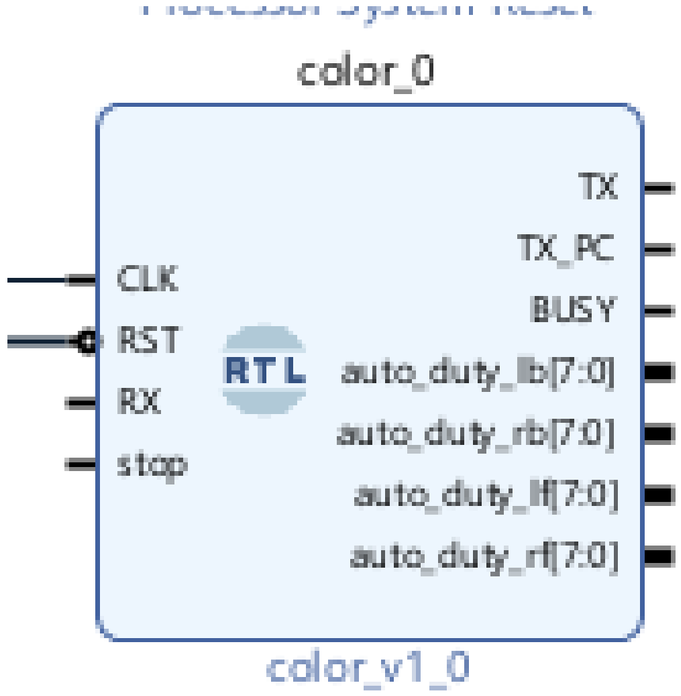
블록 다이어그램


rst_clk_wiz_5_100M
slowest_sync_clk
ext_reset_in
aux_reset_in
mb_debug_sys_rst
dcm_locked
mb_reset
bus_struct_reset[0:0]
peripheral_reset[0:0]
interconnect_aresetn[0:0]
peripheral_aresetn[0:0]
Processor System Reset
clk_wiz_1
reset
clk_in1
clk_out1
locked
Clocking Wizard
rst_clk_wiz_1_100M
Processor System Reset
buzzer_0
clk
rst
distance[15:0]
valid
buzzer
stop
RTL
buzzer_v1_0
clk_wiz
Clocking Wizard
tilt_led_3s_0
tilt_sensor
led
tilt_led_3s_v1_0
pwm_generator_0
duty[7:0]
pwm_out
pwm_generator_v1_0
rst_clk_wiz_100M
clk_wiz_5
ultrasonic_0
echo
trig
ultrasonic_v1_0
motor_pwm_uart_0
bt_rx
bt_tx
manual_duty_lf[7:0]
manual_duty_lb[7:0]
manual_duty_rf[7:0]
manual_duty_rb[7:0]
recv_data[7:0]
recv_valid
motor_pwm_uart_v1_0
color_0
CLK
RST
RX
stop
TX
TX_PC
BUSY
auto_duty_lb[7:0]
auto_duty_rb[7:0]
auto_duty_lf[7:0]
auto_duty_rf[7:0]
color_v1_0
rst_clk_wiz_4_100M
clk_wiz_4
clk_wiz_2
rst_clk_wiz_2_100M
rst_clk_wiz_3_100M
clk_wiz_3

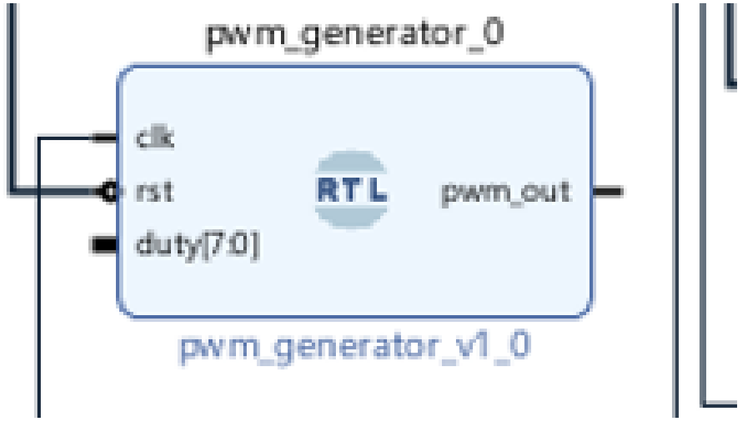
블록 다이어그램


top_0
CLK
RST
RX
BT_RX
echo
tilt_sensor
RTL
TX
BT_TX
trig
buzzer
led
TX_PC
left_fwd
left_bwd
right_fwd
right_bwd
BUSY
top_v1_0

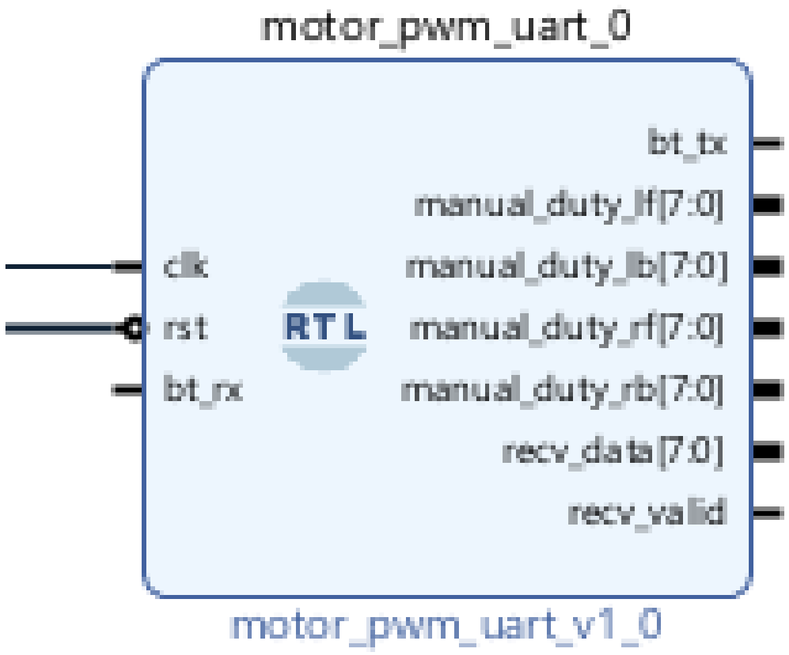
블록 다이어그램


color_0
CLK
RST
RX
stop
RTL
TX
TX_PC
BUSY
auto_duty_lb[7:0]
auto_duty_rb[7:0]
auto_duty_lf[7:0]
auto_duty_rf[7:0]
color_v1_0

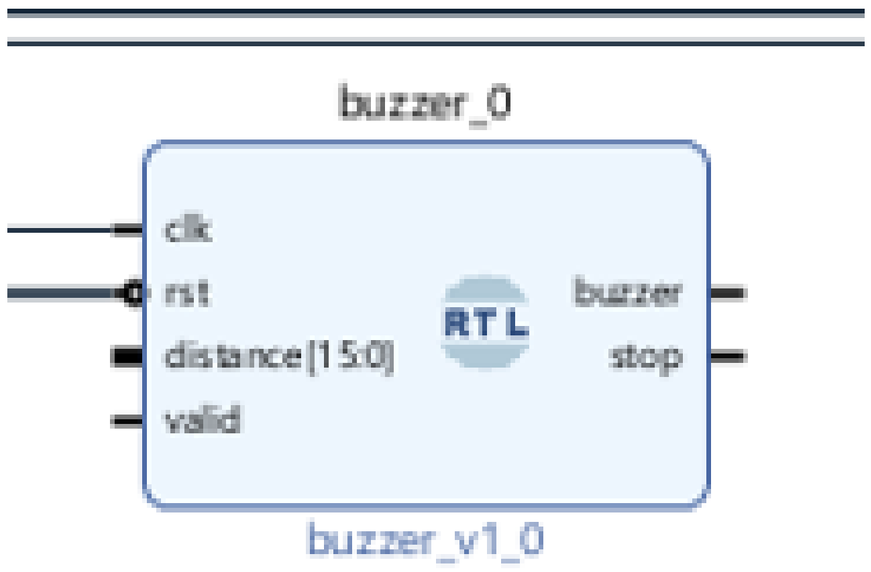
블록 다이어그램



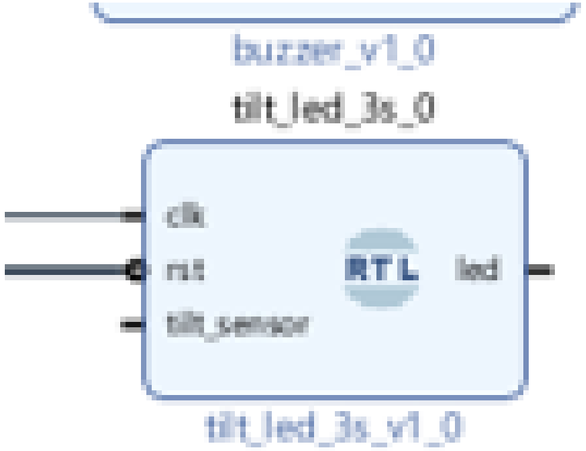
블록 다이어그램


motor_pwm_uart_0
clk
rst
bt_rx
RTL
bt_tx
manual_duty_lf[7:0]
manual_duty_lb[7:0]
manual_duty_rf[7:0]
manual_duty_rb[7:0]
recv_data[7:0]
recv_valid
motor_pwm_uart_v1_0

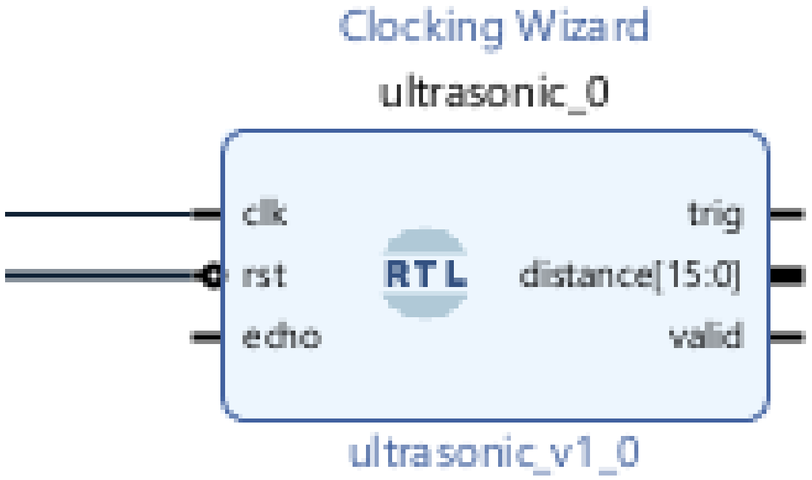
블록 다이어그램


buzzer_0
clk
rst
distance[15:0]
valid
RTL
buzzer
stop
buzzer_v1_0

블록 다이어그램


buzzer_v1_0
tilt_led_3s_0
clk
rst
tilt_sensor
RTL
led
tilt_led_3s_v1_0

블록 다이어그램


Clocking Wizard
ultrasonic_0
clk
rst
echo
RTL
trig
distance[15:0]
valid
ultrasonic_v1_0

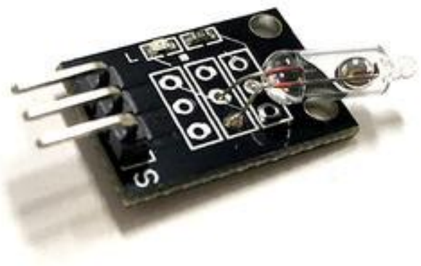
허스키 렌즈 구현 과정

1. Arrow 기반 line 트레이싱 과정

2. Color area difference를 이용한 line 트레이싱 과정

추가기능

1 기울기 센서를 이용한 LED 경고등 제작



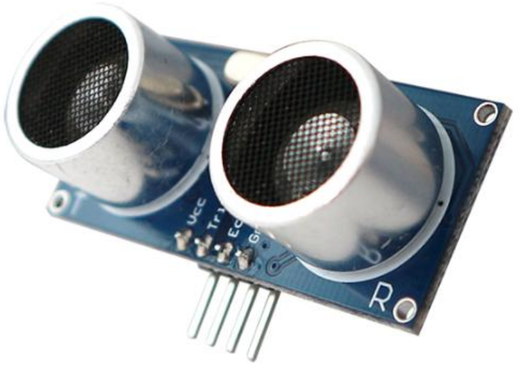
기울기 센서가 5초 이상 기울어짐을 감지

LED를 3초간 자동 점등하는 경고 시스템 설계

기울기가 복구되면 LED는 자동으로 꺼지며 시스템 초기화

추가기능

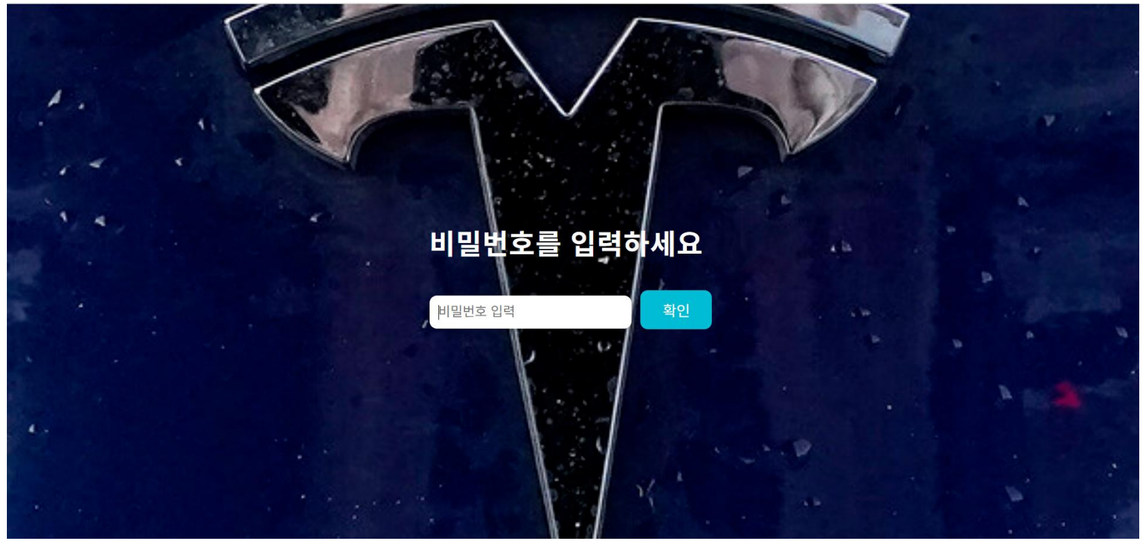
2 거리가 가까워짐(초음파)에 따라 경고음 출력



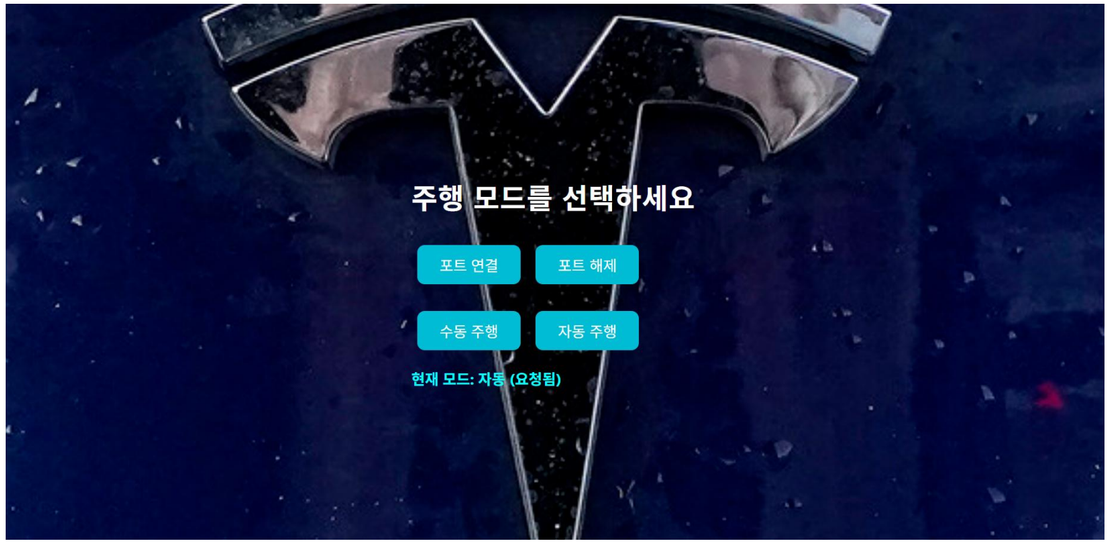
초음파 센서를 통해 거리 정보를 실시간으로 측정하고, 일정 거리 이하로 접근 시 경고음 출력

거리가 가까울수록 부저의 삐 소리가 더 빠르게 반복되며, 사용자에게 위험을 직관적으로 알림

GUI



① 비밀번호 설정을 통한 접속



② 모드 변경 및 블루투스 포트 제어

Thank you!